

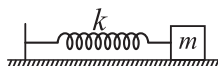
Simple Harmonic Motion

$$F = -kx$$

General equation of S.H.M. is $x = A \sin(\omega t + \phi)$; $(\omega t + \phi)$ is phase of the motion and ϕ is initial phase of the motion.

$$\omega = \frac{2\pi}{T} = 2\pi f$$

Time period (T): $T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}$



Speed: $v = \omega\sqrt{A^2 - x^2}$

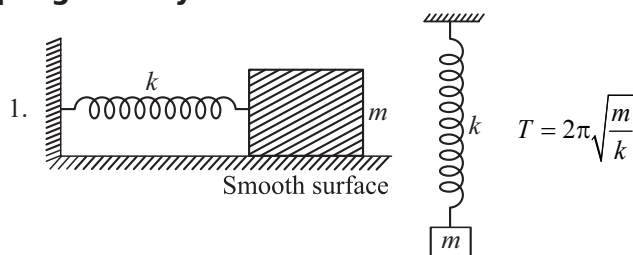
Acceleration: $a = -\omega^2 x$

Kinetic Energy (KE): $\frac{1}{2}mv^2 = \frac{1}{2}m\omega^2(A^2 - x^2) = \frac{1}{2}k(A^2 - x^2)$

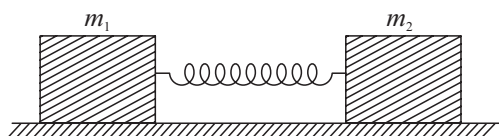
Potential Energy (PE): $\frac{1}{2}kx^2$

Total Mechanical Energy (TME)

$$= \text{K.E.} + \text{P.E.} = \frac{1}{2}k(A^2 - x^2) + \frac{1}{2}kx^2 = \frac{1}{2}kA^2 = \text{constant}$$

Spring-Mass System

2. $T = 2\pi\sqrt{\frac{\mu}{k}}$, where $\mu = \frac{m_1 m_2}{(m_1 + m_2)}$ is known as “reduced mass”.

**Combination of Springs**

Series Combination: $1/k_s = 1/k_1 + 1/k_2$

Parallel combination: $k_p = k_1 + k_2$

Simple pendulum $T = 2\pi\sqrt{\frac{l}{g}}$

$T = 2\pi\sqrt{\frac{l}{g_{\text{eff}}}}$ (in accelerating Reference Frame); g_{eff} is net

acceleration due to pseudo force and gravitational force.

Time period of simple pendulum in Accelerating lift.

(i) If velocity of lift is constant,

$$g_{\text{eff}} = g$$

$$\therefore T = 2\pi\sqrt{\frac{l}{g}} \quad (\because a = 0)$$

(ii) If lift is moving upwards with acceleration a $g_{\text{eff}} = g + a$

$$\therefore T = 2\pi\sqrt{\frac{l}{g + a}}$$

(iii) If lift is moving downwards with acceleration a ,

$$g_{\text{eff}} = g - a$$

$$T = 2\pi\sqrt{\frac{l}{g - a}}$$

(iv) If lift falls downwards freely,

$$g_{\text{eff}} = 0$$

$$T = \infty$$

Compound Pendulum/Physical Pendulum

Time Period (T): $T = 2\pi\sqrt{\frac{I}{mgl}}$

where, $I = I_{\text{cm}} + ml^2$; l is distance between point of suspension and centre of mass.

Torsional Pendulum

Time period (T): $T = 2\pi\sqrt{\frac{I}{C}}$ where, C = Torsional constant

Superposition of two SHM s along the same direction

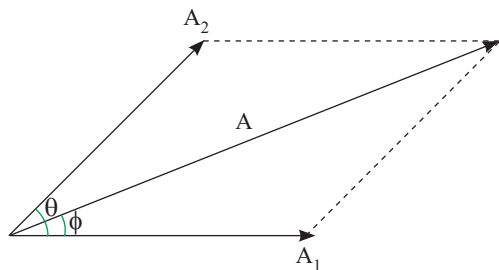
$$x_1 = A_1 \sin \omega t$$

and $x_2 = A_2 \sin(\omega t + \theta)$

If equation of resultant SHM is taken as $x = A \sin(\omega t + \phi)$

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \theta}$$

and $\tan \phi = \frac{A_2 \sin \theta}{A_1 + A_2 \cos \theta}$



Damped Harmonic Oscillations

If the damping force is given by $\vec{F}_d = -b\vec{v}$, where $\vec{v}$ is the velocity of the oscillator and b is a damping constant, then the displacement of the oscillator is given by

$$x(t) = A_0 e^{-bt/2m} \cos(\omega' t + \phi),$$

where ω' is the angular frequency of the damped oscillator, is

$$\text{given as } \omega' = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}$$

If the damping constant is small ($b \ll \sqrt{km}$), then $\omega' \approx \omega$, where

ω is the angular frequency of the undamped oscillator.

For small b , the mechanical energy E of the oscillator is given by

$$E(t) = \frac{1}{2} k A_0^2 e^{-bt/m}.$$

Forced Oscillations and Resonance

If an external driving force with angular frequency ω_d acts on an oscillating system with natural angular frequency ω_0 , the system oscillates with angular frequency ω_d . The velocity amplitude v_m of the system is greatest when

$$\omega_d = \omega,$$

a condition called **resonance**. The amplitude A_0 of the system is (approximately) greatest under this condition.